Eliminating the Parameter of Parametric Equations

Exam code: 9MA0

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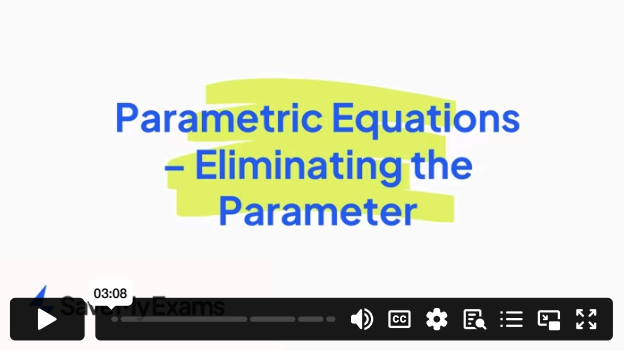

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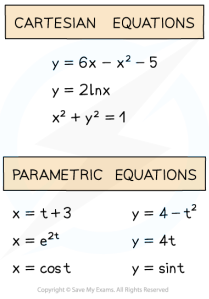
Eliminating the parameter of parametric equations

Help me understand

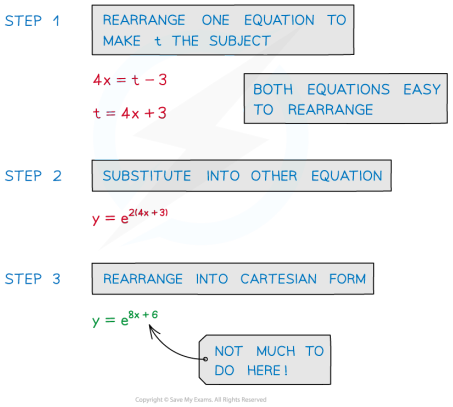
What does eliminating the parameter mean?



- In parametric equations, **x = f(t)** and **y = g(t)**
- There is still a connection directly linking **x** and **y**
  - This will be the **Cartesian** equation of the graph

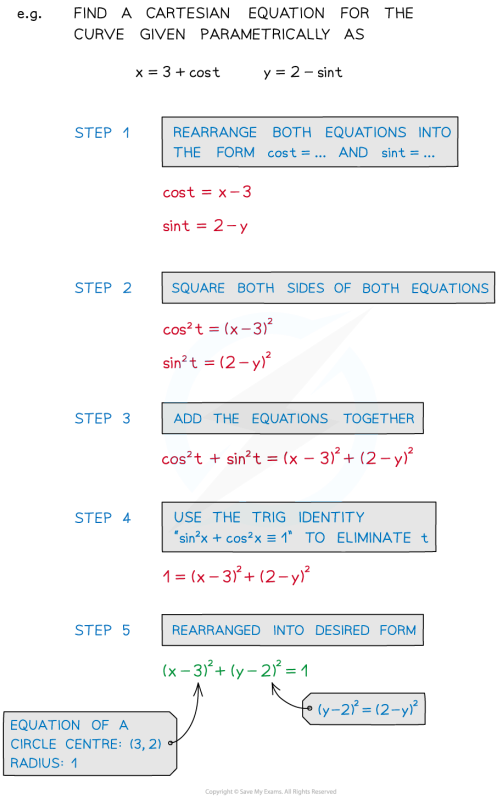
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I find the Cartesian equation from parametric equations?



- **STEP 1**
  Rearrange one of the equations to make t the subject
  - Either **t = p(x)** or **t = q(y)**
- **STEP 2**
  Substitute into the other equation
- **STEP 3**
  Rearrange into the desired (Cartesian) form

How can I eliminate the parameter using trigonometric identities?



- **STEP 1**
  Rearrange both equations into the forms "cos t = ..." and "sin t = ..."

Add the equations together

- **STEP 4**
  The trig identity "$\sin^2 x + \cos^2 x \equiv 1$" eliminates t
- **STEP 5**
  Rearrange into desired (Cartesian) form
  - This technique is seen in Trigonometric Identities

Examiner Tips and Tricks

When choosing which equation to rearrange, aim for "as simple as possible":

- Linear equations are simpler than quadratics
  - eg Rearrange $x = 2t + 3$
    or
    $y = 3t^2 + 3t - 4$?
- Single exponential terms are quite easy to deal with
  - eg $x = e^t \rightarrow t = \ln x$

Trig identities may be needed and remember **squared** terms are good!

- eg $\sin^2 x + \cos^2 x \equiv 1$

Worked Example

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